



Characterising the Variability of Black Hole Accretion Flows in GRMHD Simulations

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Abstract

In this study, we investigate the morphology and evolution of spiral-like variability structures that emerge in GRMHD simulations and present a functional form for analyzing these features directly in the image domain. We measure the shape and sizes of the spiral features, and investigate how they move and diffuse across the domain. Our analysis reveals that the pitch angle exhibits a kink that may indicate transitions in the flow dynamics. We also find evidence of a population of spiral arms with different pitch angles. We measure the spiral arms are rotating outward at around 0.03 radians per GM/c^3 . This variability model provides a new approach to characterizing variability features in GRMHD simulations. Moreover, it enables the subtraction of structured variability from both simulated and observational data. This model will also be useful to analyze variability structures in the upcoming movie campaign conducted by the EHT.

Background

- Black holes exhibit immense gravity and posses accretion flows and jets.
 - Hot-bed for extreme gravity, plasma physics, turbulence.
- Time variable structure near black hole event horizons encode information about relativistic plasma physics and accretion flow dynamics.
- The Event Horizon Telescope (EHT), through their observations have revealed structural and temporal variability on horizon scales.
- General Relativistic Magnetohydrodynamic (GRMHD) simulations provide great feasibility to quantify and study the variability structures.
 - *How can we quantify the variability structures?*

GRMHD Simulations

- GRMHD simulations solve the fluid dynamic equations of accreting plasma in Kerr space-time.
- **MAD (Magnetically Arrested Disk)** - Bumpy accretion flow and local magnetic pressure balances the inward fluid ram pressure.
- **SANE (Standard and Normal Evolution)** - Turbulent, but well ordered accretion, magnetic flux at the horizon lesser than a threshold value.
- The parameters we use for the simulation:
 - **Spin (a_*):** 0
 - **R_{high} :** 10
 - **R_{low} :** 1
 - **Inclination (i):** 163°
 - **Time of Simulation:** 25,000 – 30,000 GM/c^3
 - **Cadence:** 5 GM/c^3

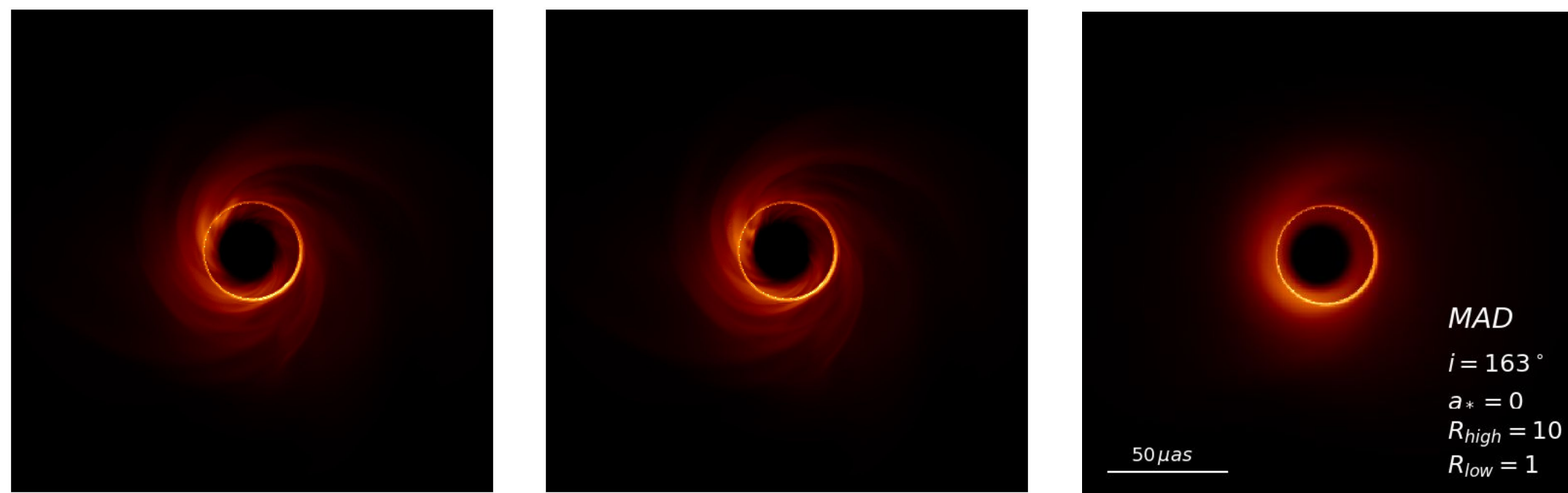


Fig. 1: GRMHD Simulation: Snapshots of the black hole M87* simulated through GRMHD for the above parameters. The first two panels represent the frames of the black hole at 5 GM/c^3 spacing. The final image is the average of 100 frames of the simulation.

This Work: Focusing on Variability

- Output of GRMHD movie - Intensity at a pixel at given time - $\xi(x, y, t)$.
- We then conduct Gaussian-smoothed temporal mean subtraction,

$$I(x, y, t) = \xi(x, y, t) - \int \xi(x, y, t) \frac{1}{2\pi T} \exp\left[-\frac{(t-t_0)^2}{2T^2}\right] dt_0$$

- We then apply the correlation based variability formulation given as,

$$C(x_1, y_1, x_2, y_2, \Delta t) = \frac{\int I_1(t) I_2(t + \Delta t) dt}{\sqrt{\int I_1^2(t) dt \int I_2^2(t + \Delta t) dt}}$$

- I_1 is intensity at pixel (x_1, y_1) and I_2 is intensity at pixel (x_2, y_2) .
- Value 1 corresponds to maximum correlation and -1 corresponds to anti-correlation.
- Follows the flip symmetry: $C(x_1, y_1, x_2, y_2, \Delta t) = C(x_2, y_2, x_1, y_1, -\Delta t)$

Application of the Formulation

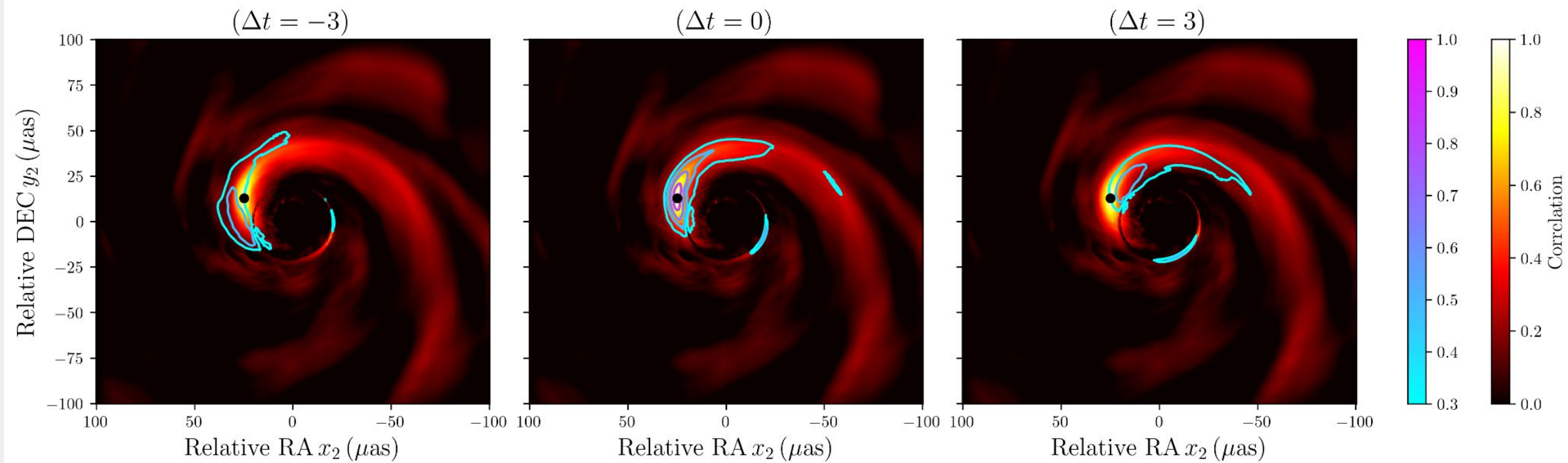


Fig. 2: Spatiotemporal Variability Structure: We apply the variability formulation to entire pixels across the image with respect to the pixel marked by the black dot. We obtain the resultant spiral arm variability as $C(r_1 = 26, \theta_1 = -1.1, r_2, \theta_2, \Delta t)$. Across the panels, the same colormap is applied, representing the variability structure at $\Delta t = 0$. The contours represent the temporal evolution. The spiral structure is moving outward clockwise as a function of Δt . We can also see a correlation signature at the photon ring on the other side due to lensing.

Converting to Polar Form

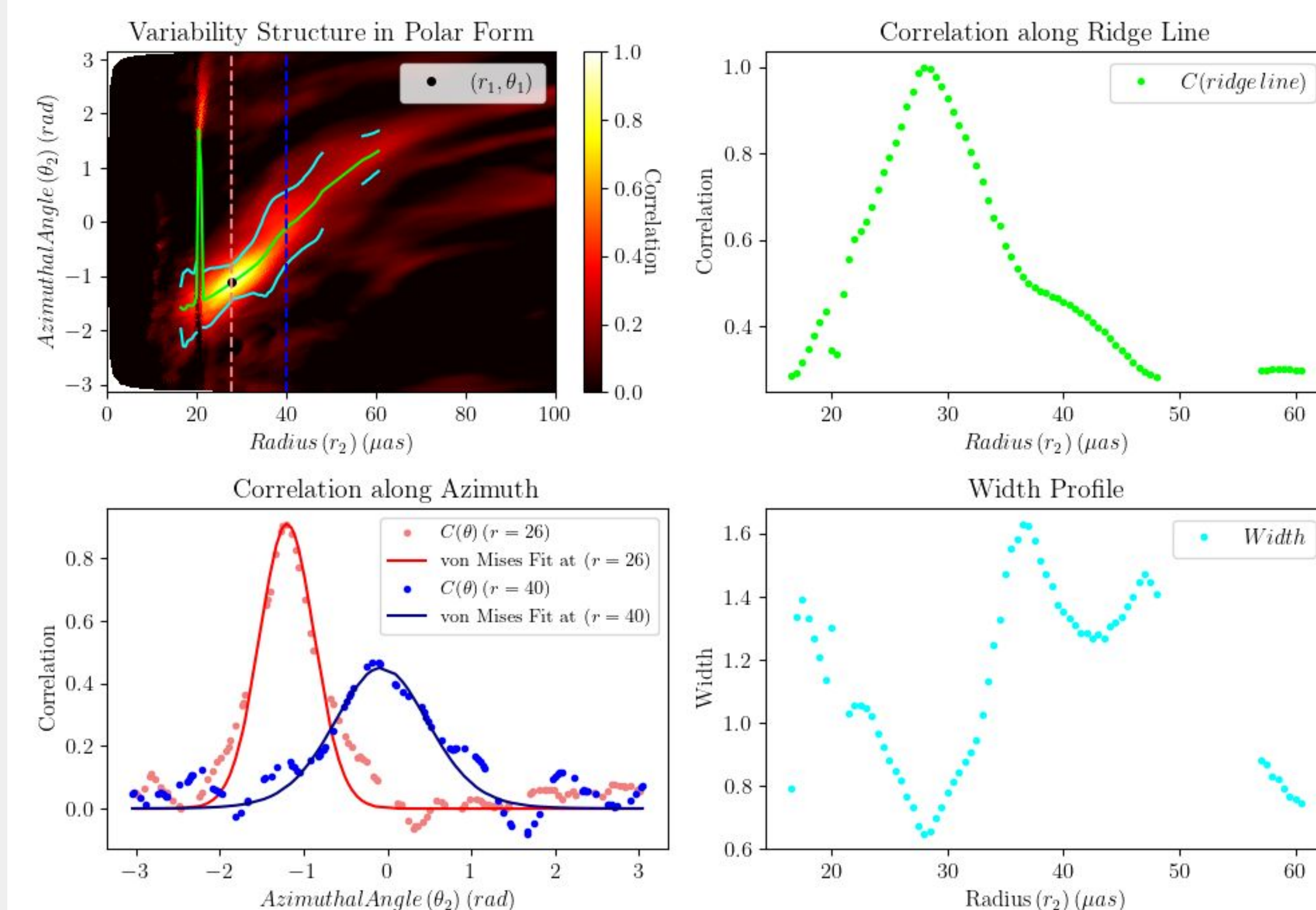


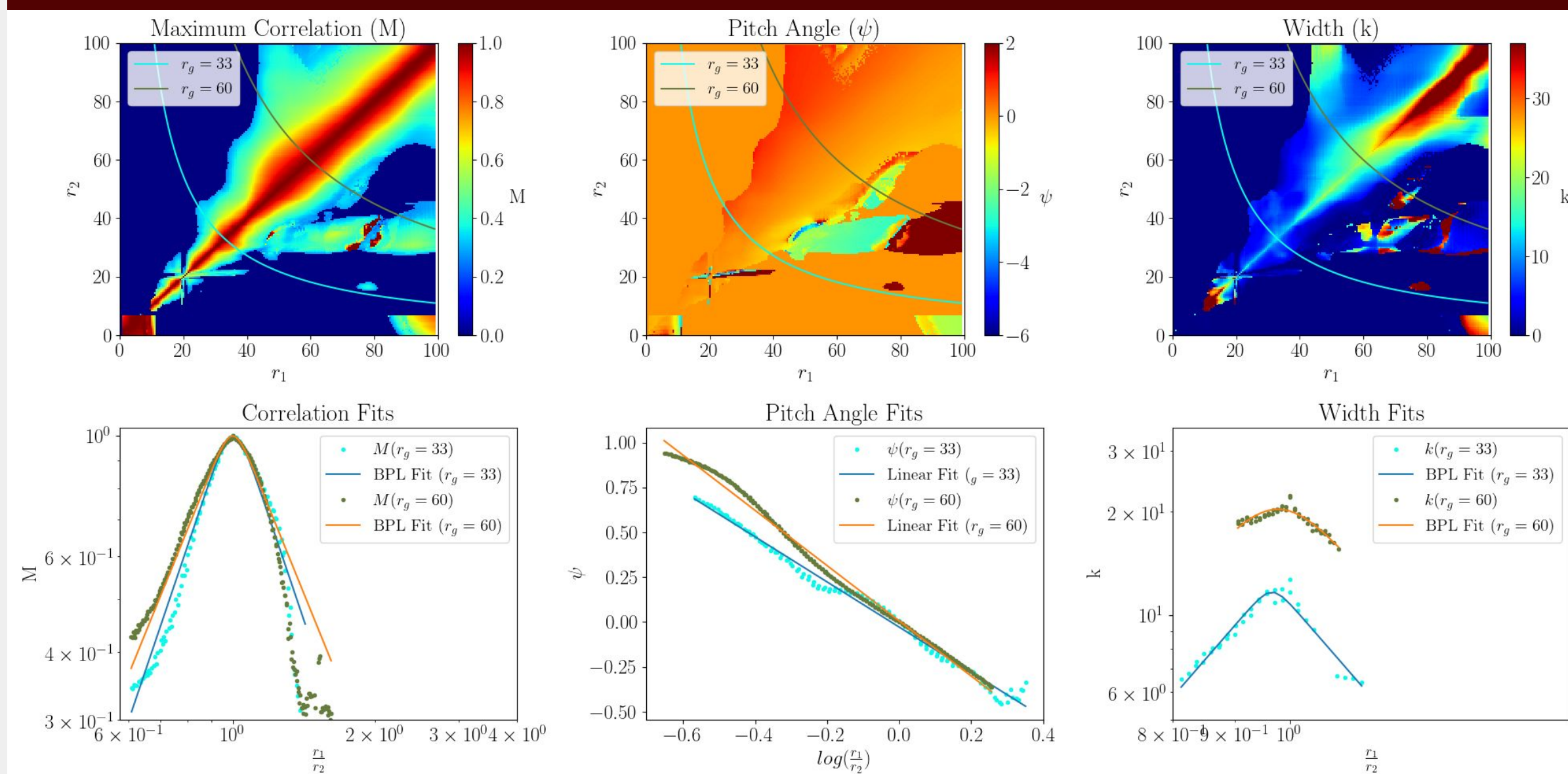
Fig. 3: In the first panel, we show the variability structure in Fig. 2 with the same reference pixels in polar form. We see that the spiral feature looks almost linear, with a kink at later radii. In the second plot, we show the correlation profile along the lime ridge line. In the third plot, we show the correlation profile in the azimuthal direction at two radial slices. We see that the structure shifts, indicating a radial dependence. In the fourth plot, we show the width profile of the feature.

The azimuthal structure of the spiral variability is modeled using a wrapped Gaussian (von-Mises) distribution:

$$C(\theta, \Delta t) = M e^{k(\cos(\theta_1 - \theta_2 - \psi) - 1)}$$

This is fit across radius and time lags to investigate spatial and temporal evolution.

Fitting Model to Parameters



Broken Power Law and Linear Approximations

Fig. 4. Broken Power Law and Linear Fits: We show the behavior of the von-Mises parameters. In the first row, we show the 2-D plots of the three parameters as a function of r_1 and r_2 . In the second row, we plot the parameter profiles and their fits of the von-Mises parameters respectively. We can that the peak correlation fit occurs at 1, the slope of the pitch angle and the value of the width change as a function of the geometric mean radius.

Broken Power Law

$$C(x) = f \left(\frac{x}{x_0} \right)^\beta \left(\frac{1}{2} \left(1 + \left(\frac{x}{x_0} \right)^{1/\Delta} \right) \right)^{-2\Delta\beta}, x = \frac{r_1}{r_2} \quad \beta(\Delta, w) = \frac{\ln 2}{2\Delta \ln \left[\cosh \left(\frac{\ln w}{2\Delta} \right) \right]}$$

Linear Model

$$\psi(x) = a(r_g, \Delta t) \log \left(\frac{r_1}{r_2} \right) + b(r_g, \Delta t) \quad r_g = \sqrt{r_1 r_2}$$

von-Mises Parameter	Symbol	Physical Significance
Maximum Correlation	M	Maximum value in the variability structure
Width	k	Width of the variability structure
Pitch Angle	ψ	How spiral is the variability structure
Broken Power Law Parameter for M	Symbol	Physical Significance
Width	w_M	Half-width half maximum of the correlation
Sharpness	Δ_M	Smoothness of the peak correlation
Location	x_{0M}	Location of the maximum value of correlation
Height	f_M	Height of the maximum value of correlation
Broken Power Law Parameter for k	Symbol	Physical Significance
Width	w_k	Radial scale of the increase in width
Sharpness	Δ_k	Smoothness of the width peak
Location	x_{0k}	Location of the maximum value of width
Height	f_k	Value of the width of the feature
Linear Parameter	Symbol	Physical Significance
Slope	a	Angle offset of the feature (Pitch Angle)
Intercept	b	Azimuthal location of the maximum correlation

Results

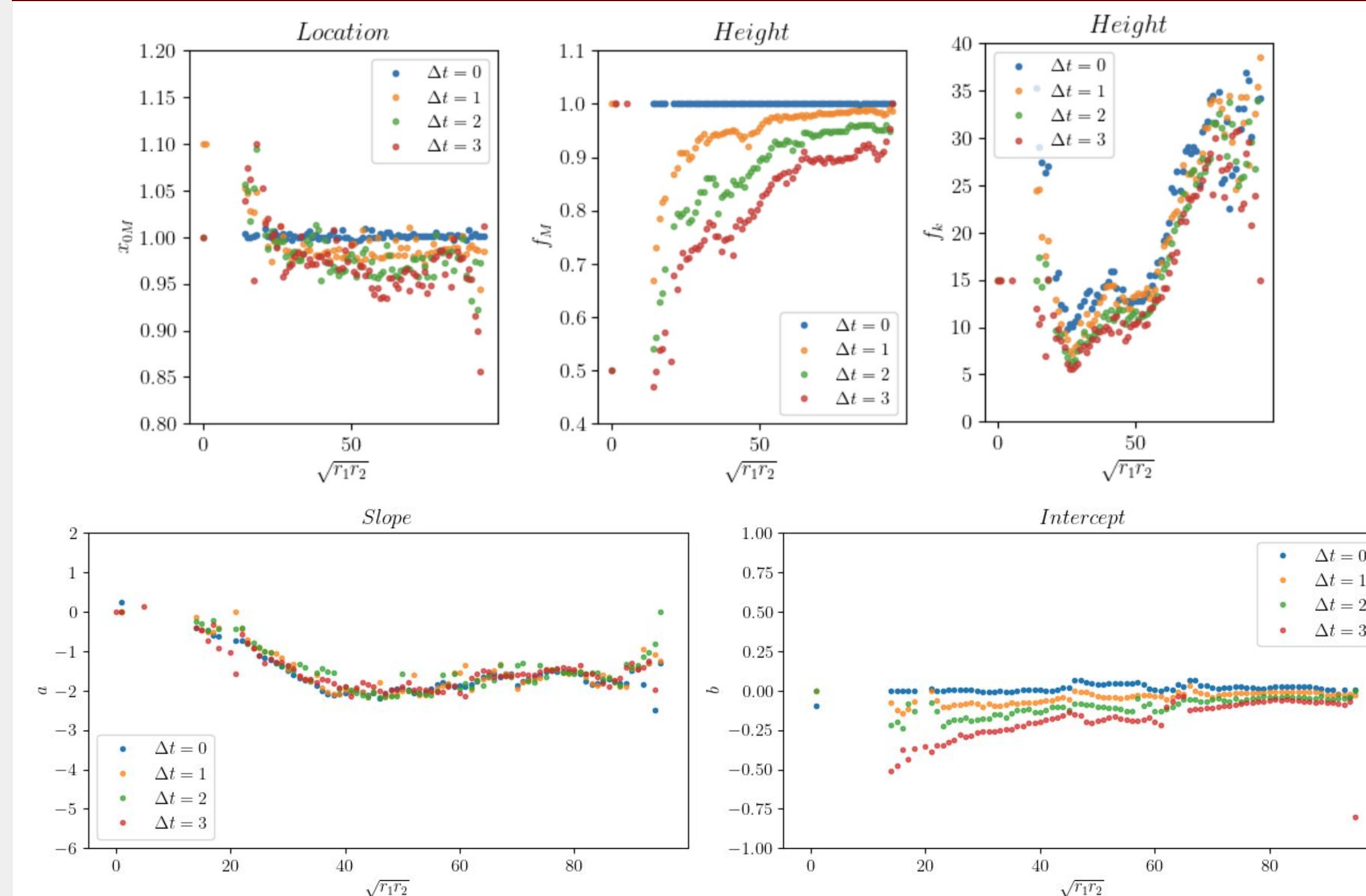


Fig. 5. Geometric Parameters: Spiral Geometry - The pitch angle varies with radius. A distinct kink appears in the pitch angle, indicating a change in flow dynamics.

Temporal Evolution - The spiral pattern shifts azimuthally with time. Measured rotation rate is 0.03 radians per GM/c^3 . Correlation decreases with time lag, indicating diffusion of variability features. **Width Parameter** - The width suggests a population of spiral arms with different pitch angles. Inner regions show thinner structures; outer regions are more diffuse.

Conclusions & Future Work

- The pitch-angle kink which likely reflects a transition in accretion flow dynamics.
- Variability spiral arms rotating clockwise.
- Population of spiral arms with different pitch angles.

Future Work:

- Extend analysis to different spins and SANE vs MAD states
- Incorporate slow-light simulations to study lensed variability effectively.

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